AIM-5014-Special-Topics-AI-Safety-and-Security (3 credits)

Adam Faulkner

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Final Project

The final project can be a case study, software application, or experiment report that deals in some way with some aspect of the course material. Case studies can include the high-profile incidents we've discussed in class, such as the OpenAI/HuggingFace incident.

The final group project deliverable is a slide deck and, optionally, a codebase implementing the group's final project.On the final day of class, groups will be given 5-10 minutes to present the results of their projects and, optionally, to demo a working implementation of their project. The presentation and optional demo ideally should be presented in-person but, if there are mitigating circumstances preventing the group from presenting in person, a pre-recorded presentation can be submitted.

Key Dates

- **Nov 13:** Project teams submit project proposals
- **Dec 9:** Project presentations

Ideas for Final Projects

1. **Case study:** *Anatomy of an AI Supply Chain Incident (Software & Package Compromise).* Conduct an in-depth retrospective analysis of a real-world or simulated major supply chain attack (such as poisoned ML dependencies, PyTorch-nightly style dependency confusion, or slopsquatting attacks targeting coding assistants). Map the attack vector using the MITRE ATLAS framework, analyze how the vulnerability propagated through developer workflows, and propose a comprehensive mitigation strategy.
2. **Case Study:** *The Regulatory and Safety Fallout of Autonomous Agent Misuse.* Examine an enterprise incident involving over-privileged tool binding or unauthorized API/web browsing actions by an AI agent (drawing on concepts from Module 3's *AI Agent Misuse &

Unauthorized Agency*). Evaluate the legal, security, and governance failures that allowed privilege escalation, and review how modern runtime guardrails or architectural isolation could have prevented the breach.

3. **Experiment Report:** *Evaluating Vector Database and Embedding Inversion Vulnerabilities*. Build a test harness or analyze existing datasets (using techniques from Module 2's privacy-preservation and embedding reversal topics) to measure how much sensitive text can be reconstructed from sentence embeddings. Compare standard vector storage against embedding-reversal-resistant techniques to quantify privacy leakage across different embedding models.

4. **Experiment Report:** *Jailbreak and Prompt Injection Robustness Benchmark.* Design a comparative data analysis pipeline testing multiple open-source LLMs against a curated suite of direct and indirect prompt injections (leveraging datasets like JailbreakBench). Measure attack success rates (ASR) before and after applying runtime guardrails.

5. **Software application:** *New AIGoat features.* AIGoat, the open-source AI red-teaming playground we've been using in class, is missing several attack types. Add 3 new attack types to the tool and demonstrate their usage live.

6. **Software application:** *Mechanistic Interpretability Dashboard for Feature Steering.* Create a lightweight interactive tool (leveraging libraries like TransformerLens) that hooks into a small transformer model to identify monosemantic features or circuits responsible for specific undesirable behaviors (e.g., sycophancy or toxic generation). The tool should allow users to apply activation addition or steering vectors in real time to observe how modifying internal activations alters model outputs.

1 Project proposal outline format

Use the following two-part structure to organize your project proposal. The proposal should not be longer than one page.

- **Names of contributors**
- **Background**
  - **For software projects:** What aspect of the AI safety and security landscape is your tool designed to address? For a red-teaming project, your background description could begin with a description of the importance of red-teaming, with a few examples, and could then describe how your tool enables users to red-team particular classes of models.
  - **For case-study projects:** What kind of AI security incident was this? Be specific: use the MITRE atlas to identify the type of incident. Why is it important to understand the specifics of the incident?
  - **For experiment report projects:** What aspect of AI safety and security does your analysis address? Motivate the analysis by describing the impact of your particular

topic to the AI safety and security landscape. For example, if you choose to present the results of experiments involving embedding reversal, explain the security implications of embedding reversal for RAG-based applications.

- **Project Outline**

  - **For software projects:** Software projects should describe the libraries and frameworks the group plans to use for their implementation. For example, if the project involves building a RAG database you could write something like, "The proposed system will use Llama2-13b as the base LLM. To index and query the data we will use the FAISS semantic search library."
  - **For case-study projects:** Provide a one-paragraph outline of what aspects of your chosen case study you plan to cover in your presentation.
  - **For experiment reports:** Provide a one-paragraph outline of the experiment set-up, data and frameworks used, and the metrics you plan to utilize when assessing experiment results